

# Notes: Garicano (JPE, 2000)

## Hierarchies and the Organization of Knowledge in Production

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# 1 Big picture

- **Question.** Why do knowledge-based hierarchies arise in production, and how do communication technology, knowledge technology, and production complexity change organizational design?
- **Setting.** Workers confront production problems. A problem must be solved for output to be produced. Workers can either learn solutions themselves or ask other workers who may know the answer.
- **Core friction.** Knowledge acquisition is costly, and communicating a problem to someone else is also costly. Because production know-how is often tacit, matching a problem to the person who knows the answer cannot be done costlessly; one has to ask.
- **Main conceptual contribution.** The paper derives hierarchy from the organization of knowledge, not from incentive problems. Supervisors exist because they hold specialized knowledge about exceptions, not because they necessarily exercise authority.
- **Main mechanism.** Production workers learn common or easy problems. Higher-level problem solvers learn rarer or harder problems. Problems move upward only when lower-level workers cannot solve them.
- **Main characterization.** The optimal organization has specialization, nonoverlapping knowledge, management by exception, vertical communication, and a pyramidal structure.
- **Key trade-off.** Adding layers economizes on knowledge acquisition because rare knowledge is held by fewer people and used when needed. But adding layers also increases communication time.
- **Technology result 1.** Cheaper *knowledge acquisition* makes workers more knowledgeable and more empowered. It increases production-worker discretion, increases span of control, reduces delay, and reduces the need for problem solvers.
- **Technology result 2.** Cheaper *communication* increases reliance on specialists. It reduces production-worker discretion but increases the span of control of problem solvers.
- **Complexity result.** A more complex or less predictable production process makes production workers rely more often on higher-level problem solvers and reduces spans of control.
- **Robustness.** Even when higher-level workers must learn everything lower-level workers know, as in on-the-job learning, the hierarchy remains specialized, pyramidal, and organized by exception.
- **Takeaway.** A hierarchy is a device for economizing on scarce knowledge and scarce communication time. Its shape depends on whether technology makes it cheaper to learn locally or cheaper to ask specialists.

## 2 Data and measurement (key objects)

## 2.1 Why this is a theory paper

- This is not an empirical paper. There is no dataset, regression table, natural experiment, or instrument.
- The paper is a formal theory of organizational design. The “measured” objects are model primitives and equilibrium objects.
- The central objects are:
  - the set of production problems,
  - workers’ knowledge sets,
  - the distribution of problems,
  - the cost of acquiring knowledge,
  - the cost of communicating knowledge,
  - the assignment of workers to layers,
  - and the allocation of problems across layers.

**Interpretation.** Read the paper as a benchmark model of hierarchy. It strips away incentives and asks: if firms only had to organize knowledge efficiently, what would a hierarchy look like?

## 2.2 Problems, solutions, and knowledge sets

Let  $\mathcal{Q} \subset \mathbb{R}_+$  denote the set of all possible production problems. A worker’s knowledge set is:

$$A \subseteq \mathcal{Q}. \tag{1}$$

A problem  $Z \in \mathcal{Q}$  is drawn. Production succeeds if the problem is in the worker’s knowledge set:

$$Z \in A. \tag{2}$$

**Interpretation.** Knowledge is modeled as a set of problems the worker knows how to solve. The model abstracts from effort and incentives and focuses on who knows what.

## 2.3 Distribution of problems

The random problem  $Z$  has distribution  $F$  and density  $f$ . The paper normalizes problems so that lower values of  $Z$  are more common:

$$f(Z) \text{ is nonincreasing in } Z. \tag{3}$$

Thus, problems near zero are common routine problems, and problems farther to the right are exceptions.

**Interpretation.** This ordering is crucial. It allows the model to say that the production floor should know common problems while higher layers should know increasingly exceptional problems.

## 2.4 Knowledge acquisition cost

Learning a set of problems costs money or time. If  $m(A)$  is the measure of the knowledge interval, then learning cost is:

$$cm(A), \tag{4}$$

where  $c > 0$  is the unit cost of acquiring knowledge.

For an interval  $[0, Z]$ , learning cost is:

$$cZ. \tag{5}$$

**Interpretation.** Knowledge has a fixed-cost flavor. Once a worker learns a solution, the knowledge can be used repeatedly, but it is costly to acquire.

## 2.5 Communication or helping cost

When a worker does not know the solution, she can ask someone else. The cost of this interaction is  $h$ .

- The cost is measured as the time of the worker being asked.
- The cost is incurred even if the asked worker does not know the answer.
- This captures the idea that determining whether a tacit problem is known is itself costly.

**Interpretation.** Communication is not just sending a message. It is diagnosing the problem, understanding whether one knows the solution, and transmitting usable instructions.

## 2.6 Organization-level objects

An organization partitions workers into  $L$  classes. For class  $i$ :

- $b_i$  is the share of workers in the class.
- $A_i$  is the knowledge set of workers in the class.
- $\ell_i$  is the list of classes that class  $i$  asks when it cannot solve a problem.
- $t_i^p$  is time spent in production.
- $t_i^h$  is time spent helping others.

The total measure of workers is normalized:

$$\sum_i b_i = 1. \tag{6}$$

**Interpretation.** The organization chooses not only how much knowledge exists but also where knowledge is located and how questions flow through the hierarchy.

## 2.7 Organizational outcomes

The model studies two broad categories of outcomes.

- **Task design:**

- how much production workers know,
- how much problem solvers know,
- and how frequently each group intervenes in production.

- **Hierarchy structure:**

- the span of control of problem solvers,
- the relative number of workers in each layer,
- the number of layers needed to solve a given share of problems,
- and the expected delay in solving problems.

**Interpretation.** The paper connects micro-level knowledge decisions to macro-level organizational shape.

## 3 Models and empirical specifications (all equations + interpretation)

### 3.1 Autarchy: one worker with no communication

Suppose one worker must solve problems alone. If the worker learns the interval  $[0, Z_a]$ , expected net output is:

$$\mathbb{E}[y] = \Pr\{Z < Z_a\} - cZ_a = \int_0^{Z_a} f(J) dJ - cZ_a. \quad (7)$$

The first-order condition is:

$$f(Z_a) - c = 0. \quad (8)$$

**Interpretation.** The worker learns solutions until the marginal probability of using the next solution equals the marginal learning cost. Common problems are worth learning; sufficiently rare problems are not.

### 3.2 Communication and organizational output

An organization allows workers to specialize in different knowledge sets and communicate when needed. For class  $i$ , helping time is:

$$b_i t_i^h = \sum_{k: i \in \ell_k} b_k t_k^p \left[ 1 - F \left( \bigcup_{\ell \prec_k i} A_\ell \right) \right] h. \quad (9)$$

Here  $\ell \prec_k i$  means that class  $\ell$  appears before class  $i$  on class  $k$ 's asking list.

Per capita output is:

$$y = \sum_{i=1}^L \left[ b_i t_i^p F \left( \bigcup_{k \in \ell_i} A_k \right) - c b_i m(A_i) \right]. \quad (10)$$

**Interpretation.** Output rises when a production problem can be solved somewhere in the relevant asking chain. Output falls with knowledge acquisition costs and communication time.

### 3.3 The organization's problem

The organization chooses:

$$\{b_i, A_i, \ell_i, t_i^p, t_i^h\}_{i=1}^L \quad (11)$$

subject to:

$$t_i^p + t_i^h \leq 1, \quad \sum_i b_i = 1. \quad (12)$$

The objective is to maximize  $y$ .

**Interpretation.** The key choice is whether many workers should learn many things, or whether rare knowledge should be concentrated in specialized problem solvers and accessed through communication.

### 3.4 Core characterization: four structural results

The model derives four core results.

1. **Specialization.** For a given knowledge allocation, workers specialize either in production or in problem solving. Only one class specializes in production; the other classes support it.
2. **Nonoverlapping knowledge.** Without an overlap constraint, no solution is learned by two different classes. Duplication wastes learning cost.
3. **Organization by frequency.** Production workers learn the most common problems. Problem solvers learn exceptions. Higher layers learn increasingly rare problems.
4. **Pyramidal organization.** Higher layers are smaller because fewer problems reach them.

**Interpretation.** Hierarchy is not imposed. It comes out of the model because the optimal assignment of knowledge is ordered by problem frequency.

### 3.5 Reduced hierarchy notation

Let level 0 be production workers. Let levels  $1, 2, \dots, L$  be problem-solving layers. Define:

$$z_i = Z_i - Z_{i-1}, \quad Z_i = \sum_{j=0}^i z_j. \quad (13)$$

Thus, production workers know  $[0, Z_0]$ , the first problem-solving layer knows  $[Z_0, Z_1]$ , and so on.

The time constraints imply:

$$b_i = [1 - F(Z_{i-1})] h b_0. \quad (14)$$

Output can be written as:

$$y = F\left(\sum_{i=0}^L z_i\right) b_0 - \sum_{i=0}^L c b_i z_i. \quad (15)$$

**Interpretation.** The size of layer  $i$  is proportional to the probability that a problem has not been solved before reaching layer  $i$ . This mechanically gives the hierarchy a pyramid shape.

### 3.6 Alternative interpretation: difficulty rather than frequency

The same logic applies if all problems are equally likely but differ in difficulty. Suppose problems are uniformly distributed on  $[0, 1]$  and the cost of learning problem  $Z$  is  $c(Z)$ , with  $c(Z)$  increasing in  $Z$ .

The cost of learning a set  $A$  is:

$$r(A) = \int_A c(Z) dZ. \quad (16)$$

Then the optimal organization has the same structure:

- production workers learn the easiest problems,
- first-level problem solvers learn harder problems,
- higher-level problem solvers learn the hardest problems,
- and the organization is pyramidal.

**Interpretation.** The economic principle is broader than frequency. Knowledge should be allocated so that commonly used or cheap-to-learn knowledge is close to production, while rare or hard knowledge is held by fewer specialists.

### 3.7 Closed-form comparative statics with exponential problems

To study technology changes, the paper assumes an exponential problem distribution:



$$F(z) = 1 - e^{-\lambda z}. \quad (17)$$

Here  $\lambda$  measures predictability. A larger  $\lambda$  means probability mass is concentrated near routine problems; a smaller  $\lambda$  means more weight on rare problems.

Under the large-organization approximation, the optimal interval length for each problem-solving layer is:

$$z_s^* = \frac{1}{\lambda} \ln \left( \frac{\lambda}{c} - \ln h \right). \quad (18)$$

The optimal interval length learned by production workers is:

$$z_w^* = \frac{1}{\lambda} \ln \left( \frac{h\lambda}{c} - h \ln h \right) = z_s^* + \frac{1}{\lambda} \ln h. \quad (19)$$

The span of control is:

$$s = \frac{\lambda}{c} - \ln h. \quad (20)$$

Equilibrium output is:

$$y^* = 1 - \frac{c}{\lambda} \left[ 1 + \ln \left( \frac{h\lambda}{c} - h \ln h \right) \right]. \quad (21)$$

**Interpretation.** Production-worker knowledge  $z_w^*$  captures discretion at the bottom. Specialist knowledge  $z_s^*$  captures the expertise of each problem-solving layer. Span  $s$  captures how many lower-level workers can be supported by one problem solver.

### 3.8 Comparative statics: communication, knowledge, and complexity

The paper's comparative statics can be summarized as follows.

**Table 1:** Main comparative statics in Garicano (2000)

| <b>Shock</b>                                    | Production-worker scope                                    | Problem-solver scope and span          | Delay / layers / intervention                                              |
|-------------------------------------------------|------------------------------------------------------------|----------------------------------------|----------------------------------------------------------------------------|
| <b>Lower communication cost <math>h</math></b>  | Decreases: workers ask more often                          | Problem-solver scope and span increase | Delay and number of layers ambiguous; problem solvers intervene more often |
| <b>Lower knowledge cost <math>c</math></b>      | Increases: workers learn more                              | Problem-solver scope and span increase | Delay falls; fewer layers are needed; problem solvers intervene less often |
| <b>Lower predictability / higher complexity</b> | Ambiguous scope, but workers solve fewer realized problems | Span decreases                         | Delay and layers increase; problem solvers intervene more often            |

**Interpretation.** Different information technologies have different organizational effects. Technologies that help workers learn make firms flatter and more decentralized. Technologies that help workers communicate can also make firms flatter, but they make production workers more dependent on specialists.

### 3.9 Overlapping knowledge extension

The baseline model assumes that higher-level knowledge need not include lower-level knowledge. The extension imposes:

$$A_i \subseteq A_{i+1}. \quad (22)$$

This captures cases in which higher-level workers must learn lower-level knowledge first, as in apprenticeship or learning by doing.

With overlap, output for an organization with  $L$  problem-solving layers is:

$$y = F(z_L)b_0 - \sum_{i=0}^L cb_i z_i. \quad (23)$$

The main results survive:

- workers still specialize in production or problem solving,
- production workers still learn common problems,
- problem solvers still handle exceptions,
- communication remains vertical,
- and the organization remains pyramidal.

**Interpretation.** Overlap creates diminishing returns to adding layers because upper layers must learn knowledge they may not use. The hierarchy remains, but the optimal number of layers is more limited.

## 4 Identification and instruments (what makes the results credible)

### 4.1 No empirical identification

- The paper does not estimate a causal effect.
- There is no instrument, no difference-in-differences design, and no randomized variation.
- Credibility comes from the internal logic of the model and from whether the assumptions map well to real organizations.

**Interpretation.** The relevant question is not “is the coefficient causal?” but “do the assumptions generate a useful and testable theory of hierarchy?”

### 4.2 What makes the model credible

The model is credible because it derives several organizational features from one simple trade-off.

- **Endogenous hierarchy.** The hierarchy is not assumed; it follows from costly knowledge and costly communication.
- **Management by exception.** The model explains why routine problems are handled low in the organization and rare problems move upward.
- **Pyramidal shape.** Higher layers are smaller because fewer unresolved problems reach them.
- **Technology predictions.** The model distinguishes communication technology from knowledge-acquisition technology and shows that they have different effects.

### 4.3 Main assumptions doing the work

- Problems can be ordered by frequency or difficulty.
- Workers know the distribution of problems.
- Knowledge acquisition has a cost proportional to the amount learned.
- Communication takes scarce time.
- Tacit knowledge means that matching a problem to an expert requires asking.
- Workers are homogeneous unless assigned to different knowledge sets.
- Incentive problems are absent.

**Interpretation.** The hierarchy is driven by knowledge economics, not by authority, monitoring, or moral hazard.

## 4.4 What the model can and cannot distinguish

- **Can explain:**

- why higher-level workers handle exceptions,
- why spans of control change with technology,
- why cheap communication and cheap knowledge acquisition have different effects,
- why more complex production needs more specialist intervention,
- and why some rare problems may be outsourced.

- **Cannot explain without extensions:**

- incentive conflicts,
- strategic withholding of knowledge,
- endogenous wages for heterogeneous worker skills,
- mistakes in referring problems upward,
- and political or authority-based aspects of hierarchy.

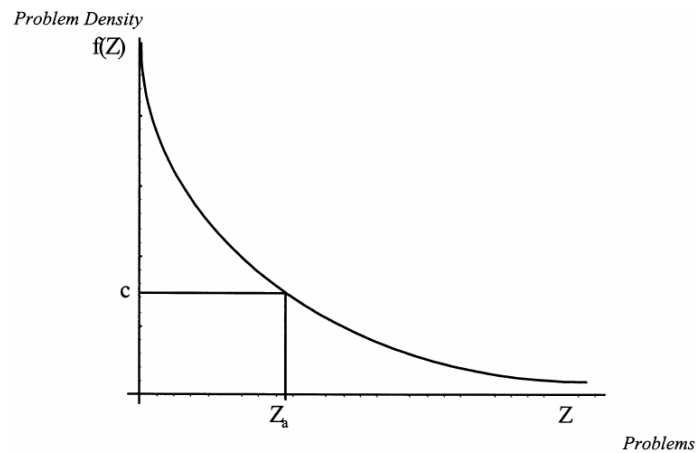
## 4.5 Relation to the literature

- Compared with incentive-based theories of hierarchy, this paper explains hierarchy from knowledge specialization rather than supervision.
- Compared with information-processing models, this paper emphasizes the acquisition and transmission of problem-solving knowledge rather than aggregation of signals.
- Compared with specialization models, it gives a concrete form to the coordination cost that limits specialization: the time cost of communicating problems and solutions.

**Interpretation.** The paper is a foundation for knowledge-based organizational economics.

## 5 Tables and figures

## 5.1 Figure 1: Knowledge acquired by a worker when communication is impossible

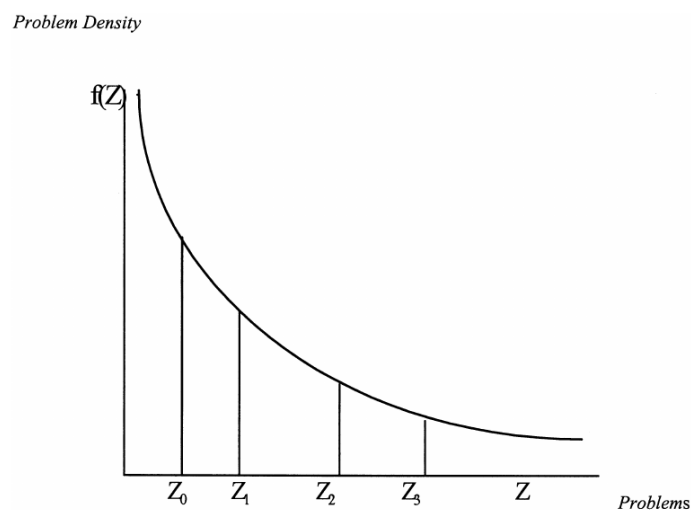


**Figure 1:** Autarchic knowledge acquisition. A worker learns problems up to  $Z_a$ , where marginal frequency equals learning cost.

**Read.** The downward-sloping curve is the problem density  $f(Z)$ . The worker learns up to the point where  $f(Z_a) = c$ .

**Economic message.** Without communication, each worker personally learns common enough problems and ignores rare problems. There is no specialization across workers.

## 5.2 Figure 2: Organization of knowledge acquisition

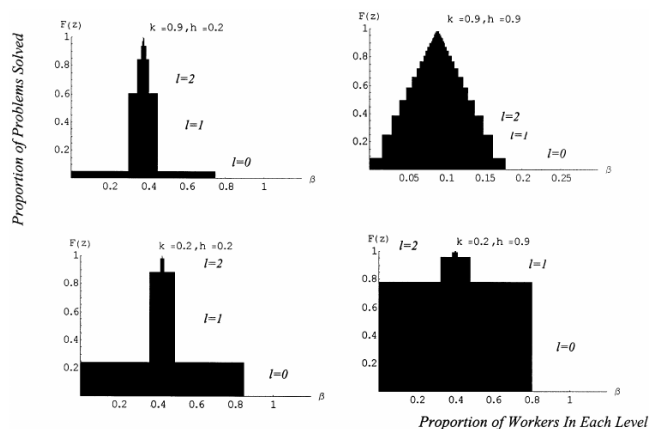


**Figure 2:** Knowledge hierarchy. Production workers learn  $[0, Z_0]$ ; higher layers learn successive intervals of rarer problems.

**Read.** The density is partitioned into intervals. The first interval belongs to production workers. Later intervals belong to progressively higher problem-solving layers.

**Economic message.** This is the core picture of management by exception. Routine knowledge is placed near production, while exceptional knowledge is placed in smaller specialist layers.

### 5.3 Figure 3: Optimal organizational forms with nonoverlapping knowledge

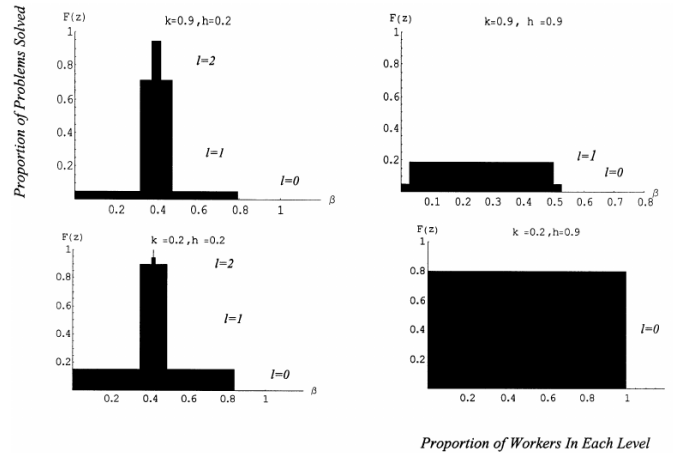


**Figure 3:** Nonoverlapping knowledge. Organizational shape varies with communication cost  $h$  and net learning cost  $k = c/\lambda$ .

**Read.** Each panel displays the share of workers in each layer and the cumulative proportion of problems solved. Moving right increases communication cost; moving up increases net learning cost.

**Economic message.** Cheap communication and cheap knowledge acquisition both increase spans of control, but they have different effects on worker discretion. Cheap communication encourages asking; cheap learning encourages solving locally.

## 5.4 Figure 4: Optimal organizational forms with overlapping knowledge



**Figure 4:** Overlapping knowledge. Higher-level workers must know what lower-level workers know, which limits the number of useful layers.

**Read.** The organization remains pyramidal, but high-cost settings have fewer layers because upper layers duplicate lower-level knowledge.

**Economic message.** Overlap does not overturn the theory. It mainly weakens the incentive to add many layers because duplicated knowledge becomes expensive.

## 5.5 Summary exhibit: propositions

**Table 2:** How to read the main propositions

| Result                  | Statement                                                         | Economic interpretation                                                                      |
|-------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>Proposition 1</b>    | Workers specialize in production or problem solving.              | A hierarchy separates doing from helping.                                                    |
| <b>Proposition 2</b>    | Knowledge does not overlap in the baseline model.                 | Duplicate learning is wasteful when higher layers do not need lower-layer knowledge.         |
| <b>Proposition 3</b>    | Knowledge is organized by frequency.                              | Common problems belong near production; exceptions belong higher up.                         |
| <b>Proposition 4</b>    | The organization is pyramidal.                                    | Fewer problems reach high layers, so fewer experts are needed.                               |
| <b>Propositions 5–7</b> | Technology and complexity reshape scope, span, delay, and layers. | Communication technology and knowledge technology have distinct organizational implications. |
| <b>Proposition 8</b>    | Overlapping knowledge preserves the same qualitative hierarchy.   | Learning-by-doing does not eliminate management by exception.                                |

## 5.6 Summary exhibit: technology comparison

**Table 3:** Communication technology versus knowledge technology

| Technology change                    | Immediate effect                     | Organizational implication                                                                   |
|--------------------------------------|--------------------------------------|----------------------------------------------------------------------------------------------|
| <b>Cheaper communication</b>         | Asking experts becomes cheaper.      | Production workers know less and ask more; problem solvers support more workers.             |
| <b>Cheaper knowledge acquisition</b> | Learning solutions becomes cheaper.  | Production workers know more and ask less; firms become flatter with more empowered workers. |
| <b>More predictable production</b>   | Routine problems become more likely. | Bottom layers solve more problems; expert intervention falls.                                |
| <b>More complex production</b>       | Rare problems become more likely.    | More specialist intervention is needed; spans of control fall.                               |

## 6 Short summary

- The paper explains hierarchy as an efficient way to organize knowledge in production.
- Workers face problems drawn from a distribution; solving them requires knowledge.
- Without communication, each worker learns common problems up to the point where marginal usefulness equals marginal learning cost.
- With communication, workers can specialize: production workers learn routine problems, and problem solvers learn exceptions.
- The optimal organization has management by exception and a pyramid shape.
- Cheaper communication makes workers rely more on specialists but increases spans of control.
- Cheaper knowledge acquisition empowers workers, reduces delay, increases spans of control, and reduces the need for layers.
- Greater production complexity increases reliance on problem solvers and reduces spans of control.
- The results survive when higher-level workers must acquire lower-level knowledge.

## 7 Conclusion

### 7.1 Core conclusion

Garicano (2000) shows that hierarchy can arise even without incentive problems. The key role of the hierarchy is to allocate knowledge efficiently when learning is costly and asking for help is costly.



## 7.2 Economic intuition

The organization wants common knowledge close to production because common problems occur often. It wants rare knowledge held by fewer workers because rare problems do not justify training everyone. This produces a hierarchy: production workers handle routine problems, and higher layers handle exceptions.

## 7.3 Takeaway

The paper's central lesson is that organizational structure depends on the relative costs of learning and communicating. If learning becomes cheaper, workers become more self-sufficient. If communication becomes cheaper, workers rely more on specialists. A hierarchy is therefore not just a chain of command; it is a technology for using knowledge efficiently.